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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

EC2221-1 – COMPUTER OPERATING SYSTEMS

(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. To access the services of the operating system, the interface is provided by the _____
A) Library B) System calls
C) Assembly instructions D) API
2. CPU scheduling is the basis of _____
A) Multiprogramming operating systems B) Larger memory sized systems
C) Multiprocessor system D) Single processor system
3. Which one of the following is not true?
A) kernel remains in the memory during the entire computer session B) kernel is made of various modules which cannot be loaded in running operating system
C) kernel is the first part of the operating system to load into memory during booting D) kernel is the program that constitutes the central core of the operating system
4. If a process fails, most operating system write the error information to a _____
A) New file B) Another running process
C) Log file D) None of the above
5. In real time operating system _____
A) process scheduling can be done only once B) all processes have the same priority
C) kernel is not required D) a task must be serviced by its deadline period
6. What are the types of distributed operating systems?
A) Zone based Operating system B) Level based Operating system
C) Network Operating system D) All of the mentioned
7. Monolithic kernel
A) is highly extensibility B) has less run time overhead
C) smaller than micro level D) suitable for real time system
8. The operating system and the other processes are protected from being modified by an already running process because _____
A) every address generated by the CPU is being checked against the relocation and limit registers B) they have a protection algorithm
C) they are in different memory spaces D) they are in different logical addresses
9. The main function of the dispatcher (the portion of the process scheduler) is:
A) swapping a process to the disk B) assigning ready process to the CPU
C) suspending some of the processes when the CPU load is high D) bring processes from the disk to the main memory

10. In multiprogramming with fixed partitions, if a process requires more memory than is available in a partition, it may lead to:
- A) Fragmentation B) Deadlock
C) Priority Inversion D) Starvation
11. A thread is also called:
- A) A virtual process B) A lightweight process
C) A heavyweight process D) A scheduler
12. If the kernel is single threaded, then any user level thread performing a blocking system call will
- A) cause the entire process to run along with the other threads B) cause the thread to block with the other threads running
C) cause the entire process to block even if the other threads are available to run D) none of the mentioned
13. Which module gives control of the CPU to the process selected by the short-term scheduler?
- A) Dispatcher B) Interrupt
C) Scheduler D) Threading
14. In _____, a programmer identifies component in a process Called _____ in a process.
- A) Paging, Pages B) Segmentation, Pages
C) Segmentation, Segments D) Segment table, Pages
15. CPU fetches the instruction from memory according to the value of _____
- A) program counter B) status register
C) instruction register D) program status word
16. _____ is generally called a translator
- A) loader B) Linker
C) Compiler or Assembler D) re-locator
17. Virtual memory allows _____
- A) execution of a process that may not be completely in memory B) a program to be smaller than the physical memory
C) a program to be larger than the secondary storage D) execution of a process without being in physical memory
18. When the page fault rate is low _____
- A) the turnaround time increases B) the effective access time increases
C) the effective access time decreases D) turnaround time & effective access time increases
19. What is FIFO algorithm?
- A) first executes the job that came in last in the queue B) first executes the job that came in first in the queue
C) first executes the job that needs minimal processor D) first executes the job that has maximum processor needs
20. Which of the following algorithms tends to minimize the process flow time?
- A) First come First served B) Shortest Job First
C) Earliest Deadline First D) Longest Job First

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Draw a neat block diagram of model of a computer system and explain the following.				
	i) Fields of Program Status Word (PSW)				
	ii) I/O subsystems and modes	8	L*2	1	1
b)	Briefly explain what batch processing system is. List the functions of Batch Monitor and with the help of timing diagram sequence, explain different components of turnaround time in batch processing systems.	8	L2	1	1

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| 2. a) | Explain goals, operation and two popular strategies of resource allocation in an Operating Systems. | 8 | L2 | 1 | 1 |
| b) | With a neat diagram, explain Kernel based operating system and also list the typical functions and services offered by the kernel-based operating system. | 8 | L2 | 1 | 1 |
| 3. a) | Define interrupts. Explain different classes of interrupts and also outline the steps involved in interrupt service routine to handle interrupts. | 8 | L2 | 1 | 1 |
| b) | Explain the structure involved in the design of microkernel based operating system with a neat diagram. | 8 | L2 | 2 | 1 |

Unit – II

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|-------|--|----|----|---|---|
| 4. a) | With a neat diagram explain the following.
i. Kernel Level threads
ii. Hybrid thread models | 10 | L2 | 3 | 1 |
| b) | Given memory partitions of 200K, 170K and 500K (in order). How would each of the First-Fit, Best-Fit and Worst-Fit algorithms place processes of 100K, 50K and 400K (in order). Explain which algorithm makes the most sufficient use of memory? | 6 | L3 | 4 | 1 |
| 5. a) | Explain fundamental state transitions for a process with neat diagram and identify five major causes of transition from running to blocked state for a program. | 8 | L2 | 3 | 1 |
| b) | Describe important features involved in process interaction between the processes. Give any two advantages of creating child processes. | 8 | L2 | 3 | 1 |
| 6. a) | Explain event handling functions of an operating system. | 10 | L2 | 3 | 1 |
| b) | Explain various steps involved in heap memory management. | 6 | L2 | 4 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Explain three preemptive scheduling policies with example. | 8 | L2 | 5 | 1 |
| b) | With a block diagram, describe the address translation in a paged virtual memory system. | 8 | L2 | 5 | 1 |
| 8. a) | Explain briefly the approaches to real time scheduling. | 8 | L2 | 5 | 1 |
| b) | What are the functions performed by virtual memory manager? Explain. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
